The Workspace Survives Where the Circuit Dies: Routing Is Not Computation

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Abstract

Anthropic's J-lens finds a global workspace in production language models: a small residual subspace carrying reusable content with a verified causal role. What the content is, and how the workspace forms, cannot be checked at scale. We run the lens inside Bayesian wind tunnels, small transformers with an analytic posterior at every position, under five preregistered predictions. Three pass: the workspace is the hypothesis frame, its contents are the posterior's surviving hypothesis set, and swapping its content redirects the model to the Bayes posterior of the swapped evidence, while the uncertainty readout riding on it is decodable and causally inert. Probing alone locates nothing: random subspaces decode nearly as well, and never redirect. The two failures matter more. The workspace is written by a bank of heads, not one. And at a loss-horizon compilation boundary the workspace survives while the computation that fills it dies: transplanting a working writer's state restores prediction from 0.19 to 1.00, and corrupting the frame-orthogonal complement destroys prediction at every depth while frame-specific damage is confined to the entry band. Routing substrate is not writer computation. The separation replicates on an LSTM and a state-space model; the frame-aligned workspace itself is the transformer's implementation, the LSTM holds the same content in a private code, and Mamba holds it in scan state, where a residual-stream positional lens cannot see it.

1 Introduction

Anthropic (2026b) report that production language models contain a "global workspace": a low-dimensional subspace of the residual stream, identified by Jacobians from internal activations to future output logits, that is (i) read from and written to far more densely than random directions, (ii) reused flexibly across tasks, and (iii) causally necessary under patching. The finding is striking and the released method is clean, but it lives where all production-scale interpretability lives: without ground truth. What the workspace *should* contain, which components *should* write it, and what its formation *should* look like are all unavailable at scale. The paper's own open question is the mechanism: what decides what enters the workspace, and how does it form?

Bayesian wind tunnels (Agarwal et al., 2026c) were built for exactly this epistemic situation. In the bijection wind tunnel, a 2.7M-parameter transformer learns in-context inversion of a random bijection; the analytic posterior at every position is known in closed form, the trained model is functionally Bayesian (entropy calibration within 0.007 bits of the posterior), and the internal mechanism is characterized: a Layer-0 hypothesis frame, elimination through middle layers, refinement late, with a derived formation account from advantage-based routing under normalized competition (Agarwal et al., 2026a). If the global workspace is a real object rather than an artifact of the lens, running the J-lens in a wind tunnel should find it, and should find it *equal to* the hypothesis frame, an identification that grounds the workspace in a setting where its contents can be checked against Bayes and supplies the formation mechanism the production-scale paper lacks.

The distinction matters. In production models, the J-lens identifies a causal subspace but cannot say what semantic object it has recovered. In the wind tunnel, where the analytic posterior is known, we can check this directly: the J-space is not merely workspace-*like*; it is the hypothesis frame carrying the surviving posterior support. What follows is not a validation of the production finding but an identification result in miniature.

We preregistered five predictions (P1–P5) with quantitative thresholds before running anything, including two designed to be falsifiable in specific directions. Three passed. The two failures are the most informative results in the note. The experiment could have failed in three qualitatively different ways: the J-space could have aligned with generic token/unembedding geometry rather than the hypothesis frame (the random-initialization and MLP controls test this); it could have decoded hypothesis identity without carrying posterior support (the P3 probes and per-hypothesis decay curves test this); or it could have patched causally without matching the Bayes posterior of the injected evidence (the P4 swaps, scored against the donor's analytic posterior, test this). The controls and swaps distinguish these cases.

Summary of findings.

1. **Identity (P1, pass).** The J-space at the embedding/block-0 band coincides with the hypothesis frame in token-identity coordinates at 4.8–6.8× a matched-dimension random-subspace null, robust to the Jacobian horizon; a random-initialization control and a trained, parameter-matched MLP control both sit at the null, so the alignment is learned structure, not unembedding geometry.

2. **Writers (P2, fail as preregistered; sharper replacement result).** No single head dominates workspace writing. Instead the nine indispensable heads (the full Layer-0 bank plus three Layer-1 heads) separate from the remaining 27 with *zero overlap* in absolute frame-projected write norm.

3. **Contents (P3, pass).** Per-hypothesis probes on J-space coordinates decode eliminated-vs-surviving status at 97.7% (r=24) and 99.8% (r=32) balanced accuracy; the rank dependence itself confirms that the workspace dimension is the hypothesis count.

4. **Causal asymmetry (P4, pass).** Swapping J-space content between evidence-matched runs redirects the model's posterior to the donor's Bayes posterior (+2.9 bits for the r=16 J-space; +8.2 bits, the full-residual ceiling, for the frame). Ablating the entropy-readout axis, which is perfectly linearly decodable (R^2=1.00), changes calibration by −0.001 bits: readable but causally inert. The frame–precision dissociation of Agarwal et al. (2026b) replicates inside the wind tunnel.

5. **The horizon boundary (P5, preregistered reverse finding).** In K=5 loss-horizon models the calibration cliff replicates (0.005–0.11 bits inside the horizon; 1.3–1.8 bits past it), but the workspace does *not* collapse: frame overlap stays at 4.8× null past the boundary while the last-layer next-token computation dies (0.77 → 0.19 decode). The compilation boundary lives in the writers, not the workspace.

6. **Cross-architecture (exploratory).** The separation principle and the frame–precision dissociation replicate on an LSTM (three seeds) and a Mamba-style SSM; the frame-aligned workspace itself does not. The LSTM holds the same causally swappable content in a private

	Preregistered claim	Threshold	Outcome
P1	J-space = frame subspace	$\geq 3\times$ null; decode $\geq 90\%$	**Pass** (4.8–6.8×; controls at null)
P2	Frame head dominates writing	$\geq 2\times$ every other head	**Fail** → 9-head bank, zero-overlap dichotomy
P3	Contents = surviving set	$\geq 95\%$ (balanced)	**Pass** (97.7% at r=24; 99.8% at r=32)
P4	Swap redirects; entropy axis inert	≥ 1 bit; calibration intact	**Pass** (+2.9 bits; ΔMAE −0.001)
P5	Workspace collapses past horizon	post-horizon $\leq$ null	**Reverse** (4.8× null; computation 0.77 → 0.19)

Table 1: Preregistration scorecard. Full thresholds, per-cell grids, and deviations are in the repository.

code; Mamba holds it in scan state, unreachable by positional patching but causally confirmed by state-level intervention (Section 4).

2 Setup

Models. The primary model is the bijection wind-tunnel transformer of Agarwal et al. (2026c): 6 layers, 6 heads, d=192, 2.69M parameters, vocabulary $2V$ with keys $0..V-1$ and values $V..2V-1$ (V=20), trained on sequences $[k_1, v_1, \ldots, k_L]$ with every key position supervised to predict the upcoming value. The trained model's entropy calibration MAE is 0.0067 bits. Controls: the same architecture at random initialization, and a parameter-matched (2.68M) attention-free residual MLP trained to its plateau (1.107 bits; it cannot bind in context, which is the point). For P5 we use the K=5 loss-horizon recurrence models of the companion manuscript (Agarwal et al., 2026d) (three seeds; modular linear recurrence $x_{t+1} = ax_t + b \bmod 17$, loss restricted to positions 1..5 of 16) and a full-horizon control.

J-lens. We follow the released operationalization (Anthropic, 2026a) with deviations recorded in the repository's `DEVIATIONS.md`, the substantive one being resolution: their estimator averages the Jacobian over source positions; the wind tunnel's mechanism is position-specific, so we keep per-(ℓ, i) resolution. For residual activation x at layer ℓ, position i, the J-space is the top-r right singular subspace of the stacked rows $\{\partial \text{logits}[p', v]/\partial x : v \in \text{hypotheses},\ p' > i\}$ over 256–512 held-out sequences, accumulated as a Gram matrix (the sweep over all layers and positions costs seconds; the entire study used ∼2 GPU-hours against a 60–120 hour budget). This is not circular: restricting the cotangent dimensions to the hypothesis tokens does not aim the lens at the frame, because the hypothesis tokens *are* the model's output vocabulary (only value positions are supervised), so the restriction is "all potential outputs," which is the released lens's own definition, not a frame-specific choice. Subspace stability across disjoint batches is 0.95 mean projection overlap (gate G0). Overlap ratios are quoted against the null mean; against the null's 99th percentile the headline 6.1× becomes 4.9×. The results are unchanged under the released estimator's own cotangent aggregation (summed over future positions: 5.9/3.6/1.8 versus 6.1/3.9/1.9 across the carrying band), so nothing rests on our per-position refinement. The comparison frame is the span of the V hypothesis-token directions as Layer-0 attention sees them (embedding mode), with the Layer-0 head-pullback variants (key-read, OV-write) reported alongside.

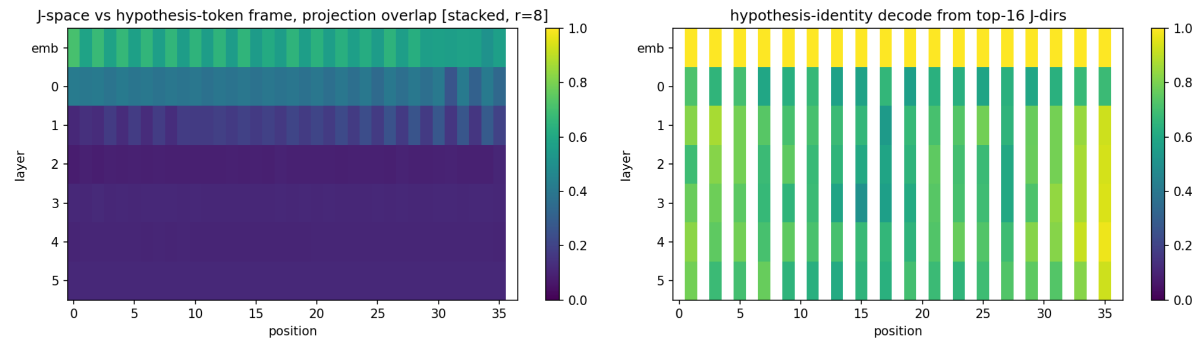
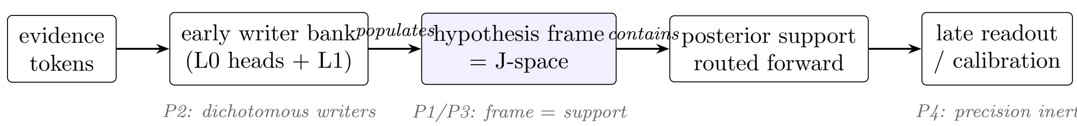


Figure 1: **P1: the J-space is the hypothesis frame.** Left: projection overlap between the per-(ℓ, i) J-space (r=8) and the hypothesis-token frame across layers (rows) and positions (columns); the carrying band is the embedding/block-0 residual. Right: hypothesis-identity decoding from the top-16 J-space directions. Late-layer cells are structurally degenerate for strictly-future targets (causality), so workspace claims are inherently early/mid-layer claims.



Past the K=5 horizon (P5): workspace geometry ✓ writer competence × readers ✓ (rescue → 1.00)
In-horizon corruption: complement × (collapse) frame: entry-band specific, generic at depth

Figure 2: The workspace pipeline as identified in the wind tunnel, with the phase that tests each stage, and the P5 dissociation: past the compilation boundary only the writers fail; in-horizon corruption places the in-flight computation in the frame's complement.

3 Results

3.1 P1: the workspace is the frame, in token coordinates

At the embedding/block-0 band the J-space overlaps the hypothesis-token frame at 4.8× (r=4), 6.1× (r=8), and 6.8× (r=16) the matched random-subspace null (Figure 1), identically for both Jacobian reductions and essentially unchanged when targets are restricted to ≥2 positions ahead (the unembedding-geometry robustness check: 4.7/5.9/6.6). The two controls are decisive (Table 2): the same architecture at random initialization gives 1.1–1.2×, and the trained MLP control gives 0.9–1.2×. The alignment is learned, and it is not the tied unembedding.

Condition	J-space/frame overlap	Rules out
trained transformer	4.8–6.8× null	—
random-init transformer	1.1–1.2×	architecture / unembedding geometry
trained MLP (2.68M, plateaued)	0.9–1.2×	training without in-context binding

Table 2: P1 controls. The frame alignment requires *trained attention*: neither the architecture's geometry nor training per se produces it.

Two refinements against the strict preregistration. First, the frame operationalization that carries the identity is the *token-identity* frame (embedding directions of the hypothesis tokens); the

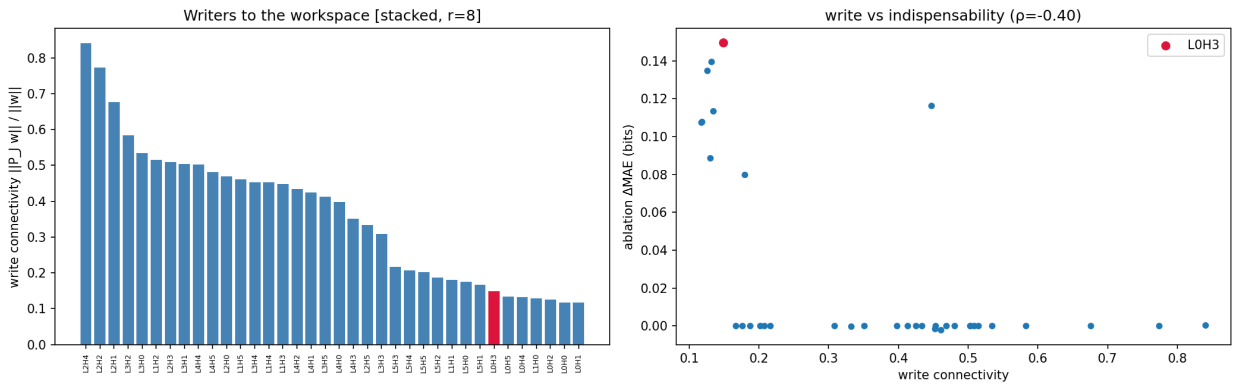


Figure 3: **P2: workspace writing is distributed but dichotomous.** Left: per-head write connectivity (ratio form); the metric is dominated by near-dead heads with tiny frame-aligned writes, which is why the preregistered ratio gate is the wrong metric. Right: write connectivity vs. ablation indispensability (ΔMAE).

Layer-0 head-pullback variants pass only at low rank (key-read 3.5–4.6× at r=4), i.e. the frame head reads the *leading* J-space directions but does not exhaust the workspace. Second, the decoding half of P1 (hypothesis identity from top-k J-space directions, 100%) is satisfied by the random-init control as well (token identity is linearly present in any embedding), so P1's evidential weight rests entirely on the trained-vs-control overlap contrast, which is 5–6×.

3.2 P2: no dominant writer, but a clean dichotomy

The preregistered prediction, that the Layer-0 hypothesis-frame head writes the workspace at $\geq$2× every other head, fails. It fails at the premise: on this checkpoint *indispensability itself is distributed*. Ablating any of the six Layer-0 heads degrades calibration by 0.11–0.15 bits (20–25× baseline), with three Layer-1 heads close behind; there is no single uniquely destructive frame head.

This is in apparent tension with the single-frame-head claim of Agarwal et al. (2026c), so we ran the reconciliation: the same ablation protocol on a paper-grade retrain of Paper I's task variant (shared vocabulary, explicit query token; calibration 0.017 bits). There the claim replicates: one Layer-0 head stands out at 2.7× the second, with a single head above half the top effect. The production variant (no query token; every key position is a readout site) instead distributes indispensability across the full Layer-0 bank. Writer concentration tracks readout demand: one query site, one binding head; a readout site at every position, a bank. The single-head claim is variant-true, not architecture-general, and the two results are one phenomenon seen at two supervision geometries.

The replacement result is sharper than the prediction it replaces. Ranking heads by *absolute frame-projected write norm* $\|P_F\, w_h\|$, the nine indispensable heads (all of Layer 0 plus L1H0/H1/H3) separate from the remaining 27 with zero overlap: minimum 2.06 among the indispensable, maximum 0.75 among the rest. Writer magnitude into the workspace *is* indispensability, as a dichotomy (Spearman $\rho = 0.47$, $p = 0.004$ on the continuous ranking; the within-group ordering carries no signal). A methodological caution for the production-scale analysis falls out of the failure: the scale-free ratio $\|P_J w\|/\|w\|$ is gamed by near-dead heads whose tiny writes happen to point at the workspace (ratio-form $\rho = -0.40$ with indispensability, wrong sign); write connectivity should be measured in absolute projected norm.

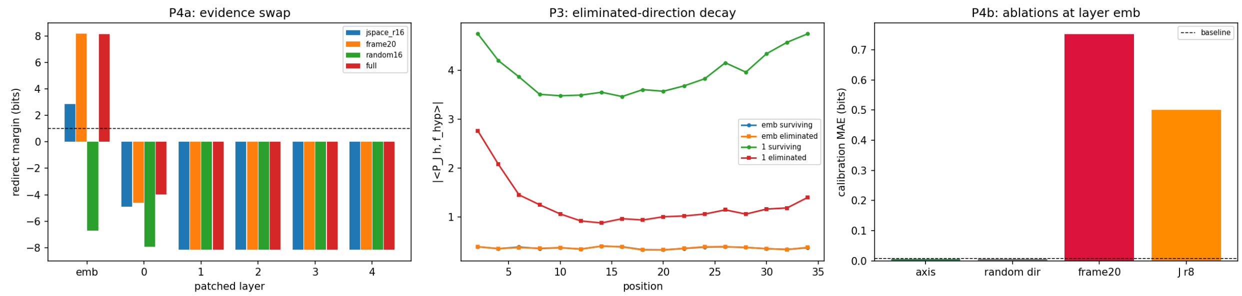


Figure 4: **P4: causal structure of the workspace.** Left: evidence-swap redirect margins by patched layer; the frame swap at the embedding equals the full-residual ceiling, the random-subspace control never redirects, and the injection window closes after layer 1. Center: eliminated-hypothesis directions decay in J-space-projected residual norm as elimination proceeds. Right: ablations; the entropy axis is causally inert, the frame is not.

3.3 P3 + P4: contents carry the posterior support; the model uses the content, not the precision readout

Contents (P3). Per-hypothesis logistic probes on J-space coordinates decode eliminated-vs-surviving status at 90.0% (r=16), 94.8% (r=20), 97.7% (r=24), 99.8% (r=32) balanced accuracy (full-residual ceiling 100%). The monotone rank dependence crosses the preregistered 95% exactly as the probe rank passes the hypothesis count V=20. But a referee control cuts the probe evidence down to size: random rank-matched subspaces decode at 0.96 (r=24) and 0.99 (r=32), nearly matching the J-space. Decodable information is promiscuous in the residual stream, so the probe form of P3 does not locate content; it shows only that the J-space carries what most fat subspaces also carry. This is our own readable-is-not-used caution applied to ourselves, and it moves the content-location claim entirely onto the causal test below, where the discrimination is total: the J-space swap redirects (+6.5 bits at matched rank r=20) and no random subspace ever does (mean −11.5, maximum −10.9 over 100 draws). (Preregistration note: the eliminated base rate approaches 0.9 late in the sequence, so raw accuracy is uninformative and the gate was evaluated with balanced accuracy.)

Causal asymmetry (P4). We build evidence-matched donor pairs (identical key order, independent permutations) and patch the J-space component of the evidence region with the donor's. At the embedding, the r=16 J-space swap redirects the model's posterior toward the *donor's* Bayes posterior by +2.9 bits (margin over the original posterior, 256 pairs), and +6.5 bits at the rank-matched r=20; the 20-dimensional frame swap redirects by +8.2 bits, indistinguishable from patching the entire residual; random matched-dimension subspaces never redirect (mean −11.5, maximum −10.9 over 100 draws). Content routed through the workspace is what the model treats as evidence.

The dissociation half: the entropy-readout axis (fit by ridge regression; $R^2 = 1.00$, perfectly linearly decodable) changes calibration by −0.001 bits when projected out. Projecting out the frame costs +0.75 bits. Being readable and being used are different properties, measured causally — and the lens agrees: the entropy axis's projection onto the top J-directions is at or below the random-direction baseline at every layer, so the use-weighted subspace excludes what the ablation shows to be inert; the frame–precision boundary result of Agarwal et al. (2026b) replicates in a 2.7M-parameter model. This also discharges a standing question from our own backlog: probes

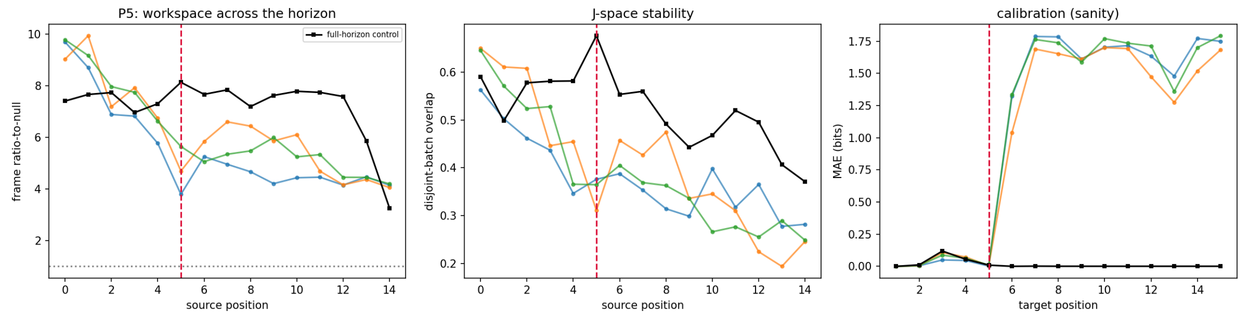


Figure 5: **P5 reverses: the workspace survives the compilation boundary.** Left: frame overlap ratio-to-null by source position for three K=5 loss-horizon seeds (colors) and the full-horizon control (black); the dashed line marks the horizon. The preregistered collapse to null does not occur. Center: J-space stability. Right: the calibration cliff replicates exactly.

on the (a, b)-style readout directions detect real structure, and that structure can still be causally epiphenomenal. The precise statement: the workspace *carries posterior support*; precision is readable off it but not necessarily used.

A mechanistic bonus. The swap experiment doubles as a causal map of *when* evidence enters the readout pathway: patches at block outputs ≥ 1 have no effect at all (margins pinned at the unpatched baseline, for every basis including the full residual). The injection window closes after layer 1, exactly where P2 found every indispensable writer. Two independent methods, one localization.

3.4 P5: the reverse finding at the horizon boundary

The K=5 loss-horizon models replicate the companion manuscript's compilation cliff precisely: calibration MAE is 0.005–0.11 bits at trained positions and 1.3–1.8 bits past them, against 0.001 everywhere for the full-horizon control. The preregistered prediction was that the workspace tracks the cliff: well-formed at positions ≤ 5, degrading to null past the boundary. The preregistration also stated that the reverse outcome "must be reported if found." It was found (Figure 5).

Past the horizon the frame-aligned J-space persists at 4.8× null (a decline from 6.5× in-horizon, but nowhere near the $\leq 1.5\times$ collapse gate; two independent full-horizon control seeds hold 6.7–7.7 in-horizon and 6.7–7.1 past it). What collapses is the *writer computation*: decoding the (deterministic) next token from J-space coordinates at the last layer succeeds at 0.77 inside the horizon and falls to 0.19 past it, while the control holds 1.00 everywhere. The dissociation is exact: geometry global, writer computation positional.

We read this as the strongest bridge result between the two papers. The loss-horizon wall of Agarwal et al. (2026d) is real and replicates at the circuit level. But the wall is in the *writers*, the positionally compiled machinery that fills the workspace with calibrated content, not in the workspace itself. Conversely, the flexibility Anthropic attributes to the workspace is a property of the routing substrate and does not, by itself, buy generalization past a compilation boundary: a model can carry a perfectly good workspace into positions where nothing knows how to write to it.

Horizon rescue by writer patching. The dissociation admits a direct causal test: if the boundary lives in the writers and not in what follows them, then handing a post-horizon position the state an in-horizon writer would have produced should rescue the prediction. Modular orbits make the required state-matching exact: the donor sequence is the test sequence's orbit started six steps later, so donor position p carries the recurrence state of test position $p+6$. We transplant the donor's residual at in-horizon positions 2–4 into post-horizon positions 8–10. The rescue is complete and monotone in patch depth: post-horizon accuracy goes from 0.19 unpatched to 0.78 (block 1), 0.98 (block 2), and 1.00 (block 3), across all three seeds, while the full-horizon control is unharmed by every patch condition. (The writer span here extends through block 3; in the bijection model the injection window closed after layer 1. These are different tasks with different write depths; the modular-inverse computation is deeper than key–value binding.) The readers, everything downstream of block 3 at that position, are position-general; given a correctly written state they finish the computation perfectly, anywhere. Two controls sharpen this. Transplanting only the frame-subspace component does *not* rescue (0.21, indistinguishable from a random-subspace transplant at 0.24): mid-stream, the in-flight (a, b) computation lives in the residual complement of the token frame, rotating into frame coordinates only late. The corruption sweep below sharpens this: in the recurrence model the frame's causal role is confined to the entry band, and the in-flight and readout computation live in the complement. The apparent tension with P4 is a difference between models and bands, not a contradiction: in the bijection model the frame swap at the embedding is the full-residual ceiling; here, past entry, the frame carries nothing the computation uses. And patching at the embedding does nothing (0.20) because the matched states share their input token; everything the donor contributes is computed structure from the writer layers. The compilation boundary is now causally localized: writers (blocks 0–3 at the position) are broken past the horizon, readers are not.

Writer corruption: the converse. The loop closes from the other direction, with a referee-requested control that changed our reading. At *in*-horizon positions, where everything works, we corrupt the residual with norm-preserving random orthogonal rotations at every block (1–5), in four conditions: the full residual, the frame's orthogonal complement, the frame itself, and a random subspace of the frame's dimension. Complement rotation collapses prediction at every depth (1.00 → 0.05–0.36; chance 0.06); frame rotation spares it at blocks 3–5 (0.98–1.00), but so does the dimension-matched random rotation (1.00 everywhere), so sparing at depth is a statement about dimension, not about the frame. The frame-specific signal is at the entry band: at block 1, frame rotation damages prediction (0.38–0.54) while the random matched-dimension rotation does nothing (1.00), across all three seeds and the control. The frame is causally specific where evidence enters, fades to generic by block 3 (the same migration the rescue's depth profile shows), and the complement carries the in-flight computation throughout. Every condition is norm-matched; the subspace-specific causal discrimination lives in the swap and rescue contrasts, which are dimension-matched by construction.

Why the workspace must survive. The reverse finding is a structural consequence of weight sharing. The objects that define the workspace geometry, the token embeddings and the QK/OV maps, are position-independent by construction; position enters only through how those shared weights are *used*, via content routed by attention. Gradient descent under a positional loss mask can therefore compile position-dependence only into the computation routed through the shared weights, never into the weights themselves: the geometry is global because nothing in the parameterization can make it local. "Geometry global, writer computation positional" is then the expected signature of positional compilation in any weight-shared architecture, which is the thesis of Agarwal et al. (2026d), now visible through the lens. (And yes, this is "just weight sharing"; that is the point.)

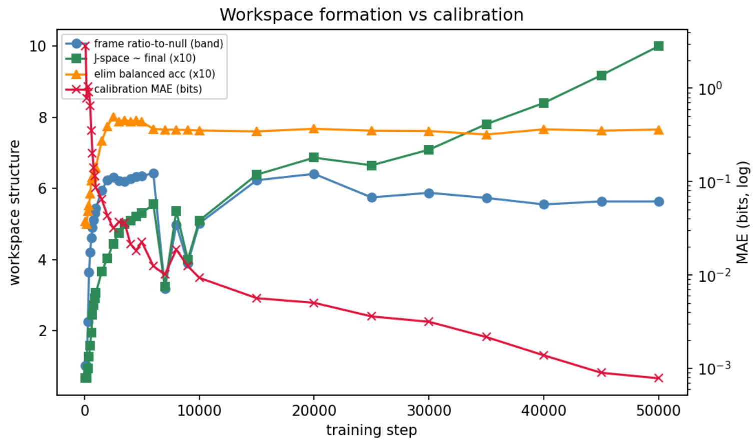


Figure 6: **Workspace before precision** (dense-checkpoint retrain, steps 100–50k, log-scale MAE). The frame ratio crosses the P1 threshold by step 400 and plateaus by step ∼2,500; calibration MAE improves another ∼40× after the frame is in place.

Dose–response. The checkpoint family includes runs trained with sparse post-horizon supervision (`extra_loss_density` $\in \{1\%, 5\%, 25\%\}$ of post-horizon positions receiving loss), which convert the binary reverse finding into a graded law. Across densities 0% → 1% → 5% → 25% (three seeds each): post-horizon calibration MAE 1.64 → 0.54 → 0.014 → 0.005 bits and last-layer decode 0.19 → 0.68 → 1.00 → 1.00, while the workspace overlap ratio stays high throughout (4.8 → 4.8 → 7.6 → 7.4× null). Supervision density is the variable, writer competence is what it buys, and the workspace is the thing it does not need to buy: even 1% post-horizon supervision recovers most writer function against an already-present substrate.

3.5 Formation dynamics

A retrain with dense early checkpoints (same architecture, data distribution, and hyperparameters, reimplemented single-GPU from the production trainer; checkpoints every 100 steps from step 100) makes the formation curve visible end to end (Figure 6). The retrain converges somewhat further than the production checkpoint (0.0008 vs 0.0067 bits final MAE, plausibly a distributed-vs-single-GPU trainer difference); the formation *ordering*, which is the claim, is unaffected. The frame rises fast and first: overlap with the frame subspace is 2.3× null at step 300, crosses the P1 threshold (3×) by step 400, and plateaus near 6× by step 2,500, at which point calibration MAE is still 0.03 bits and proceeds to improve another ∼40× (to 0.0008 bits) over the remaining 47k steps with the frame ratio flat. Elimination decoding saturates on the same early schedule (0.54 at step 300, 0.79 by step 3k). The production run's checkpoints (10k–50k) show the same plateau from above: already 3.7× at its first checkpoint, converged to the final workspace by 30k. This is the ordering the frame–precision dissociation predicts: build the frame, then sharpen the numbers in it.

The early workspace is causally functional. Formation and causation sit on one x-axis in Figure 7: rerunning the P4 evidence-matched frame swap at every dense checkpoint, the redirect margin crosses the +1-bit success threshold between steps 400 and 500 (+0.84 → +1.92 bits), the same steps at which the frame crosses the P1 overlap threshold, while calibration is still 0.65 bits, three orders of magnitude from converged. The margin then grows monotonically with training (+3.4 at step 600, +5.4 at 1k, +11.5 at 50k). Norm-preserving corruption at block 0 damages

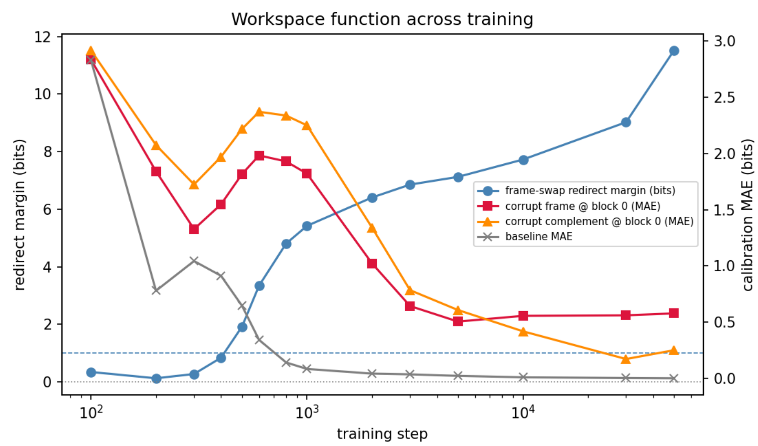


Figure 7: **The early workspace already does causal work.** Per-checkpoint P4 frame-swap redirect margin (left axis) against calibration under frame/complement corruption at block 0 (right axis, log-x). The margin crosses the +1-bit causal threshold between steps 400–500, as the frame crosses the P1 threshold, and grows monotonically thereafter.

calibration from the first functional checkpoint (metrics follow the task: accuracy on the recurrence task, where the target is deterministic; calibration MAE here, where it is a posterior), and its structure sharpens with training: early, frame- and complement-rotation hurt about equally; by step 30k frame-rotation is 2.3× more destructive (+0.58 vs +0.25 bits), the load consolidating into the frame coordinates at the injection band, consistent with P4's frame-ablation result. Two migration axes, one object. Across depth within a forward pass, content moves frame → complement → frame; across training time, load consolidates into the frame at the injection band. The formation story is therefore causal end to end: the workspace forms by step ∼400, becomes causally functional as soon as it forms, and the remaining 47k steps buy writer precision inside a substrate that already works.

4 Beyond transformers

Everything so far is one architecture. Is the workspace a fact about trained Bayesian systems, or about transformers? We ran the same pipeline (identity, contents, swaps, ablations) on an LSTM (6-layer, 1.80M; three seeds) and a Mamba-style selective SSM (1.45M; two seeds for identity, one for interventions) trained on the identical task. These runs are exploratory (post-preregistration), parameter counts are not matched, and one property of the task matters for reading them: with keys drawn without replacement and no repeated-key queries, Bayes-optimal calibration requires only *accumulation* (tracking spent values), the one primitive the taxonomy of Agarwal et al. (2026c) grants every recurrent architecture. All three architectures accordingly calibrate (transformer 0.007, LSTM 0.009–0.018, Mamba 0.012–0.013 bits). That is what makes the comparison informative: same Bayesian function, three substrate organizations (Figure 8).

The separation principle is architecture-general. In every calibrated architecture, the J-lens finds a causal workspace whose contents decode the posterior's surviving set (transformer 0.998; LSTM 1.000 at layer 3–4, r=24, all three seeds; Mamba 0.965), and in every one the entropy-readout axis is near-perfectly decodable and causally inert (ablation ΔMAE $\leq$ 0.0002 bits in all three). The frame–precision dissociation is not a transformer fact; it is a trained-Bayesian-system fact.

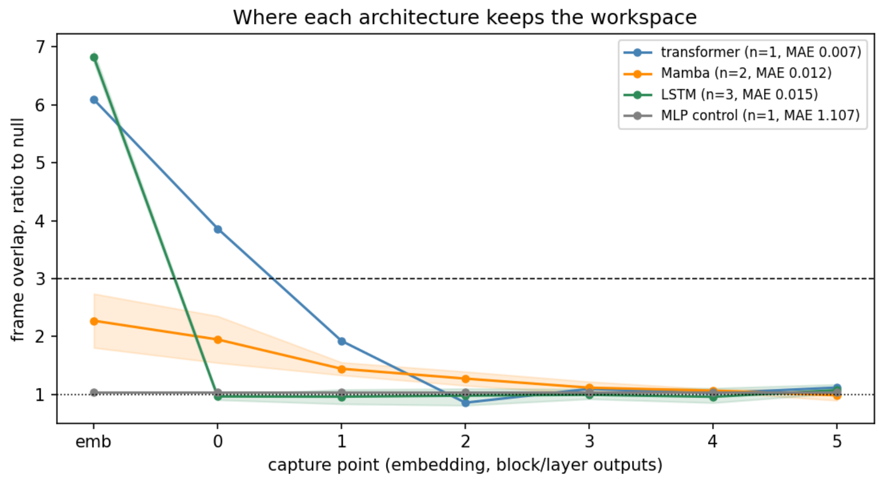


Figure 8: **Where each architecture keeps the workspace.** Frame overlap (ratio to null, r=8) by capture point; bands span seeds. The transformer holds a frame-aligned workspace in its residual stream (emb–block 1); the LSTM is frame-aligned *only at the embedding* (6.7–6.9×, three seeds) and exactly null through all recurrent layers; Mamba is graded; the MLP control is null everywhere.

The coordinates are architecture-specific: the LSTM's private code. The LSTM presents a clean dissociation of its own. Its recurrent layers sit at exactly null frame overlap, and frame coordinates there are causally dead: projecting out the frame at hidden layers costs nothing (0.009 → 0.009 bits) and frame-component swaps have no effect (−6.8 bits margin). Yet the same layers carry the full posterior content in the LSTM's *own* J-space coordinates: probes decode the surviving set perfectly, and swapping those coordinates between evidence-matched runs redirects the posterior by +3.5 to +4.9 bits (three seeds). At the embedding boundary, where evidence enters in token coordinates, the frame swap equals the full-residual ceiling (+6.6 to +6.9 bits), after which the content is re-encoded into a private code. Same content, same causal structure, different coordinate system.

Mamba: the workspace leaves the residual stream entirely. On the SSM, *no* positional patch redirects the posterior: not the frame, not its own J-space, not the full residual of the entire evidence region (best margin −0.6 bits). This is not unpatchability but the wrong lever: evidence transport is distributed through per-block scan states rather than positional residuals. Intervening on the states themselves settles necessity robustly: zeroing every block's state at the read boundary is catastrophic on both seeds tested (KL to the true posterior explodes from ∼0.02 to 25–34 bits; the states carry everything). Sufficiency by transplant is not seed-stable: an evidence-matched *state* swap redirects the posterior on one seed (+1.2 bits at the deep read position, halfway at mid-sequence where local convolution, skip, and gating paths carry recent evidence) and fails on a second (−2.3), plausibly because a selective SSM's donor states are trajectory-specific in ways positional residuals are not. What stands: the evidence lives in scan state, off the residual stream, where a residual-stream positional lens cannot see it.

Reading. The separation principle (routing substrate versus writer computation) holds in all three architectures. What is architecture-specific is the *implementation of the routing side*: attention keeps the substrate as a frame-aligned positional subspace of the residual stream (its content-addressable keys require token coordinates); the LSTM keeps the same content at layer boundaries in a learned private code; Mamba keeps it in scan state, off the residual stream altogether. A practical implication for production interpretability follows: the J-lens finds workspaces where architectures keep content in positional residual subspaces, so on hybrid or SSM-based models the same analysis could report "no workspace" while an intact, causally verified one sits in state space. One caveat bounds the reading: this task exercises accumulation only; on a variant requiring binding (repeated-key queries), the recurrent architectures are expected to fail outright (Agarwal et al., 2026c), and whether their private-coordinate substrates persist into the failure regime is an open question for the follow-up.

5 Discussion

The contribution is one sentence: *routing substrate is not writer computation.* A workspace can exist, be correctly shaped, and be causally available while the machinery that fills it is absent. The identification results are what license this separation principle: without P1/P3 grounding what the workspace *is*, "routing geometry is not computation" would be a claim about an uninterpreted subspace. And the distinction is orthogonal to whether a workspace exists: a model may possess an intact routing substrate yet fail to generalize because the computations that populate it have not been learned.

What is now grounded. In this wind tunnel the J-space is identified with the hypothesis frame (P1), it carries exactly the posterior's support (P3), its content is what patching moves (P4), and the formation mechanism the production paper asks for is the one already derived for the frame: advantage-based routing builds it early, a bank of early heads writes it, and precision arrives later (P2, dynamics). None of this could be verified at production scale; all of it survives verification here.

What changed under test. Two preregistered predictions failed in ways that revise the picture rather than merely bounding it. Workspace writing is distributed but dichotomous, not concentrated. The correct writer metric is absolute projected write, a caution that transfers directly to the production analysis. It also bears on low-rank adaptation: the task's functional subspace has dimension near the hypothesis count (probes saturate once rank covers it), which is the activation-side version of the premise LoRA rests on, and it yields two testable corollaries. Adapters on writer layers should move content while adapters placed late should move precision, and adapter sites selected by relative norms are exposed to the same anti-selection failure measured here. And the workspace does not inherit the compilation boundary: position-bound failure lives in the writers, now causally localized by the rescue experiment (readers are position-general; a transplanted writer state restores post-horizon prediction to 1.00). In one line: *workspace existence is not workspace competence.* Stated as architecture: the workspace is a routing substrate, and generalization depends on the learned writers, not on the existence of the substrate itself. If this transfers to scale, "the model has a workspace" and "the model can use its workspace here" are separate claims requiring separate evidence, a distinction the production paper's flexibility experiments do not currently draw.

Relation to the independent reception. An early independent review of the production release (Nanda, 2026) replicated its core claims on a second model family and set the practitioner's calibration: the lens is an approximation, best treated as hypothesis generation; expected to miss some content and to carry false positives; and least trusted where it is used causally. The wind tunnel speaks to all three, in both directions. On causal use: scored against the analytic posterior, the lens-identified subspace redirects exactly as Bayes requires (P4). On false positives: the most tempting candidate in our setting, the perfectly decodable entropy axis, is not in the J-space — its projection onto the top J-directions sits at or below the random-direction baseline at every layer (for example 0.10 against an expected 0.20 at r=8, block 1) — so the use-based definition resists exactly the artifact that probing invites. On false negatives: we exhibit one causally verified instance, the state-resident workspace of Section 4, and name its mechanism. The review's implicit missing experiment, validation on a task with known ground truth, is this note.

Relation to the frame–precision dissociation. The separation principle has a genealogy inside this program. The frame–precision dissociation of Papers I–III separates two kinds of *content* within a working circuit: the frame carrying posterior support, and the precision readout riding on it, replicated causally here by P4b (the entropy axis is perfectly decodable, $R^2 = 1.00$, and causally inert, -0.001 bits). Routing-vs-computation is the same schema one level up: not support versus confidence within content but *substrate versus process*: the coordinate system versus the machinery that writes into it. The learning dynamics carry the same signature at both levels: geometry first, precision second (Papers I–II); substrate causally functional by step $\sim$400, writer precision refined over the remaining 47k steps (Figure 7), with the load consolidating into frame coordinates as training proceeds, plausibly the same EM-like sculpting described in Paper II. The promotion is strict: P5 exhibits the substrate existing with the process entirely absent, a configuration the frame–precision work had no setting to produce. The genealogy in one sentence: frame–precision was the first cut of the separation; the loss-horizon models supplied the setting in which the two objects fully come apart; the J-lens supplied the instrument that measures the substrate independently of the process.

Relation to identifiability theory. The J-lens has a formal cousin in the nonlinear-ICA identifiability literature, which studies when latent structure is recoverable from the Jacobian of a generative map (Hyvärinen and Pajunen, 1999; Zheng et al., 2022; Zheng and Zhang, 2023; Zheng et al., 2026). The objects differ: that line proves guarantees for the Jacobian of the map from latent sources to observations, under invertibility and structural sparsity or diversity assumptions, while the J-lens probes the Jacobian of a trained network's forward pass from activations to logits, where those assumptions do not hold and no guarantee is claimed. Two of its results nonetheless bear directly on ours. First, identifiability in that setting holds only up to permutation and component-wise transformation of the latents; our finding that the token-identity frame carries the P1 overlap while head-pullback variants pass only at low rank is recovery up to exactly such a transformation class, the expected indeterminacy rather than a defect of the preregistration. Second, the stated frontier of that literature is checking generalized identifiability in trained foundation models, where ground-truth latents are unavailable. The wind tunnel is the setting where they are available: P1 and P3 jointly verify, in a model with an analytic posterior, that the functional subspace the Jacobian identifies equals the generative latent structure it should identify. To our knowledge this is the first empirical case where the two notions can be computed independently and compared, and they coincide.

The one-sentence version. Bayesian computation factorizes into a routing substrate and writer computation. The separation is architecture-general; the geometric realization of the substrate is architecture-specific. In these models, the transformer externalizes it into a shared token frame, the LSTM internalizes it into a private code, and the SSM moves it into scan state. That is the claim the rest of the paper supports.

Limitations. Wind-tunnel scale is the point, and the caveat: these are 2.7M-parameter models with one in-context task each, and the late-layer J-space is structurally degenerate for strictly-future targets, so depth claims are confined to early/mid layers. The identification threshold ($3\times$ null) was preregistered but the frame mode that carries it (token-identity rather than head-pullback) was a within-family refinement; we report all modes. Read-connectivity ratios here are 2–$3.6\times$, far from the $\sim 100\times$ reported at production scale, consistent with a 6-layer model having little room for specialization, but unverified. The hypothesis space is single-token by construction, so the production-scale concern about concepts without single-token equivalents is untested here.

Artifacts. Code, per-cell grids, preregistration deviations, and all figures: `experiments/jlens/` in the wind-tunnel repository, pinned against `anthropics/jacobian-lens @581d3986`. The full study is reproducible in $\sim$2 GPU-hours on one consumer GPU.

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